

Augmented IPW and Doubly Robust Estimators

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Advantages/Disadvantages of IPW versus regression model

- Both approaches require assuming consistency and no unmeasured confounding
- IPW requires assuming positivity
- IPW requires correctly specifying propensity model; regression approach requires specifying the outcome model
- IPW: can fit a single model (propensity score) and then obtain causal estimates for several outcomes
- Perception that modeling treatment allocation “easier” than outcome
- IPW tends to be more variably (even without extreme weights)

Augmented IPW estimators

- If we take the point of view that the propensity score model is correctly specified, then one can use semiparametric theory to show that all consistent and asymptotically normal estimators for $E(Y^1)$ can be written as

$$\hat{E}(Y^1) = \frac{1}{n} \sum_{i=1}^n \frac{A_i Y_i}{\pi(X_i; \hat{\gamma})} - \{A_i - \pi(X_i; \hat{\gamma})\} h(X_i) \quad (1)$$

- where $h(X_i)$ is any arbitrary function of X
- Show that this is a consistent estimator of $E(Y^1)$ regardless of the choice of $h(X)$
- This class of estimators is known as augmented IPW estimators
- Questions: how to choose $h(X)$

Choosing $h(X)$

- Although all these estimators are consistent for $E(Y^1)$ they will have different variability
- One can show using semiparametric theory that the choice that leads to the estimator with the smallest asymptotic variance is

$$h(X_i) = \frac{E(Y|A = 1, X)}{\pi(X)} \quad (2)$$

- Let's think through the implication of this choice. Note that $E(Y|A = 1, X) = E(Y^1|X)$ under our standard identifying assumptions
- ① When $A_i = 0$ (i.e., Y_i^1 is unobserved) then the summand in equation (1) is $E(Y_i^1|X_i)$ – effectively we are imputing the best guess of the potential outcome given covariates
- ② When $A_i = 1$ (i.e., Y_i^1 is observed) then the summand is $E(Y_i^1|X_i) + \frac{1}{\pi(X_i)}(Y_i^1 - E(Y_i^1|X_i))$

Choosing $h(X)$

- Showing that

$$h(X_i) = \frac{E(Y|A = 1, X)}{\pi(X)} \quad (3)$$

leads to the estimator with the smallest asymptotic variance is
DIFFICULT

- But showing this choice of $h(X)$ leads to an estimator with smaller variance than the standard IPW estimator (i.e., $h(X) = 0$) is more doable

Choosing $h(X)$

- Of course, $E(Y|A, X)$ is not known to us (if it were, we would have just used regression modeling to estimate $E(Y^1)$)
- Nonetheless, just like regression modeling, we can posit a model for $E(Y|A, X) = \mu(A, X; \eta)$ and then obtain an estimate for η (say $\hat{\eta}$)
- We can then plug this estimate into the formula so that the estimator is

$$\hat{E}(Y^1) = \frac{1}{n} \sum_{i=1}^n \frac{A_i Y_i}{\pi(X_i; \hat{\gamma})} - \frac{\{A_i - \pi(X_i; \hat{\gamma})\}}{\pi(X_i; \hat{\gamma})} \mu(A = 1, X_i; \hat{\eta}) \quad (4)$$

Doubly Robust Estimator

$$\hat{E}(Y^1) = \frac{1}{n} \sum_{i=1}^n \frac{A_i Y_i}{\pi(X_i; \hat{\gamma})} - \frac{\{A_i - \pi(X_i; \hat{\gamma})\}}{\pi(X_i; \hat{\gamma})} \mu(A = 1, X_i; \hat{\eta}) \quad (5)$$

- Consistent and asymptotically normal if either $\mu(A, X; \eta)$ or $\pi(X; \gamma)$ correctly specified
- Known as doubly robust or doubly protected estimator
- Note that if $\mu(A, X; \eta)$ is misspecified, the augmented estimator is no longer guaranteed to have smaller variance than the “standard” IPW estimator
- Some refer to summand as pseudo-outcome

Estimator for $E(Y^0)$

- If we take the point of view that the propensity score model is correctly specified, then one can use semiparametric theory to show that all consistent and asymptotically normal estimators for $E(Y^0)$ can be written as

$$\hat{E}(Y^0) = \frac{1}{n} \sum_{i=1}^n \frac{(1 - A_i)Y_i}{1 - \pi(X_i; \hat{\gamma})} - \{1 - A_i - (1 - \pi(X_i; \hat{\gamma}))\}h(X_i) \quad (6)$$

- The optimal choice of $h(X_i)$ is then $h(X_i) = \frac{E(Y|A=0,X)}{1-\pi(X)}$

Form Pseudo-Outcomes for AIPW

```
P01 <- imai$VOTED98*w1 - ((imai$PHN.C1-ps)/ps)*pred1  
P00 <- imai$VOTED98*w0 - ((1-imai$PHN.C1-(1-ps))/(1-ps))*pred0  
ATE_AIPW <- mean(P01 - P00)
```

Voting Example: AIPW Analysis

```
[1] "Average Treatment Effect"
```

```
[1] 0.102
```

```
[1] "Bootstrap SE"
```

```
[1] 0.033
```

```
[1] "Bootstrap Normal 95% CI"
```

```
[1] 0.0377 0.1672
```

AIPW Results: Key Assumptions

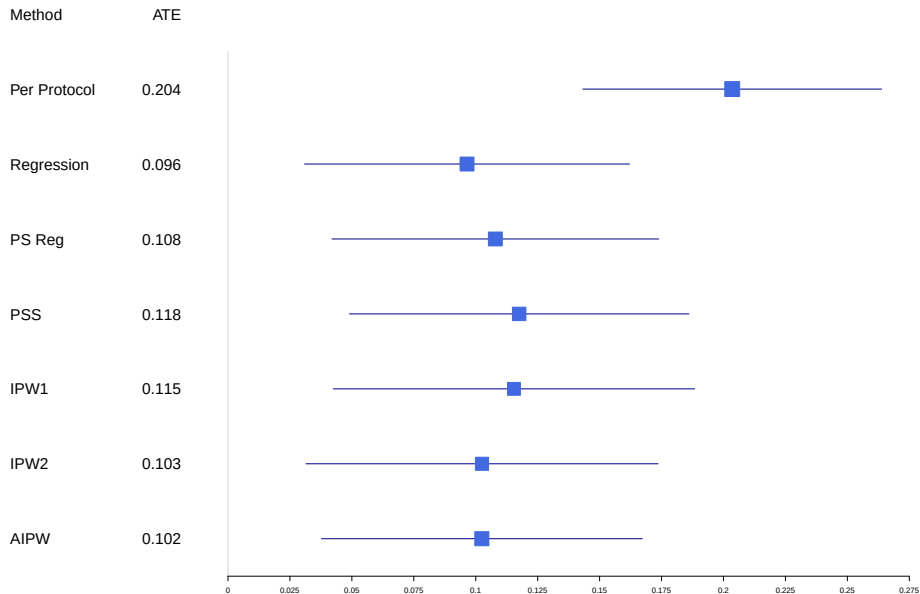
Identifying

- ① Consistency
- ② No Unmeasured confounding
- ③ Positivity

Modeling

- ① Propensity score model (given all confounders) OR outcome model (given all confounders) correctly specified.

Putting it All Together



Simulation Example

- Let $X_i \sim N(0, 1)$, $Y_i^1|X_i \sim N(0.5 + \gamma X_i, 1)$ and $Y_i^0|X_i \sim N(\gamma X_i, 1)$. $ATE = E(Y^1) - E(Y^0) = 0.5$
- Note that in the “real world” we do not observe $\{Y_i^1, Y_i^0\}$ but would observe $Y_i = A_i Y_i^1 + (1 - A_i) Y_i^0$. This implies that $Y_i|A_i, X_i \sim (0.5A_i + \gamma X_i, 1)$
- Let $A_i|X_i \sim \text{Bernoulli}(p_i)$ where $p_i = \exp(0 + \alpha X_i) / \{1 + \exp(0 + \alpha X_i)\}$
- Generate a sample of size 500 consistent with this data generating mechanism with $\gamma = 1$ and $\alpha = 1$.
- With these coefficients the R^2 for regressing the outcome on X in placebo group is 0.5 and C-index for the treatment allocation ≈ 0.75
- 100 Monte Carlo datasets

Simulation Results

	bias	sd
IPW	-0.00575	0.136
REG	0.00381	0.109
AIPW	0.00084	0.116